

The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by a thin black border, with dark gray horizontal bars at the top and bottom.

ORACLE

Academy

Database Foundations

5-1

Mapping Entities and Attributes

ORACLE
Academy



Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

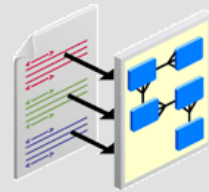
Objectives

- This lesson covers the following objectives:
 - Explain the naming conventions used in a relational database.
 - Use Oracle SQL Developer Data Modeler to create a glossary and apply naming standards to
 - Map entities to table names
 - Map attributes to column names



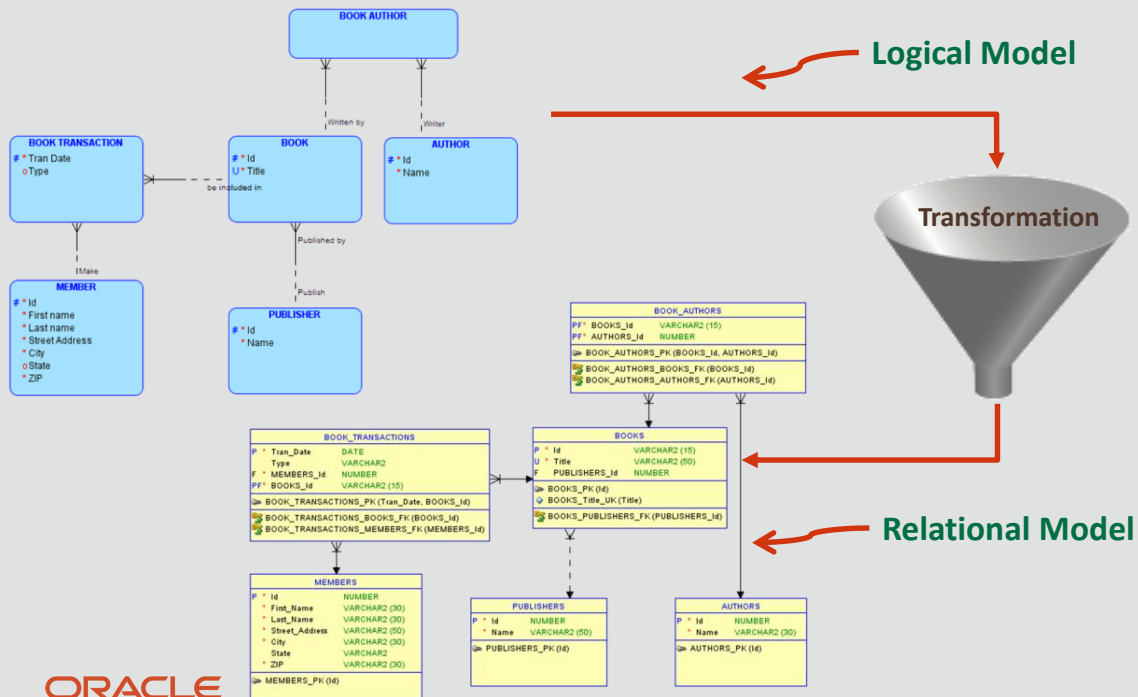
Need for Creating a Physical Model

- A Physical model:
 - Is the blue print for the actual database implementation
 - Can be used as the basis for implementing any type of DBMS
 - The ideal model can be adapted to an RDBMS model
 - In Oracle SQL Developer Data Modeler a Physical model is known as a Relational Model



An Entity Relationship Model (ERM) does not highlight the physical and database constraints. It is essential to transform the ERM into a relational model that can serve as the basis for defining the physical implementation of the database.

Transformation Process: Logical to Physical (Relational) Model



ORACLE
Academy

DFo 5-1
Mapping Entities and Attributes

Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

5

The logical model (ERD) captures the requirements for a business through graphical elements such as entities, relationships, and arcs. This model is a generic representation and does not describe the complexities involved in physical implementation.

This means that, when you try to fit in all of the syntax from a logical model (ERD) directly into the physical model, you may miss out on specific business rules. To preserve these specified rules and keep track of them, you need to transform the logical model into a relational model.

Naming Conventions

- Decide on conventions for:
 - Table names:
 - The plural of the entity name is generally used as the corresponding table name
 - Column names and special characters (% , * , # , - , space , ...)
 - Column names are often identical to the attribute names, with a few exceptions. Replace special characters with an underscore because SQL does not allow most special characters to be used in column names. Often column names are abbreviated
 - Table short names (abbreviations)
 - A unique abbreviation for every table is a very useful element for the naming of foreign key columns or foreign key constraints

The naming of constraints is covered in the next lesson.

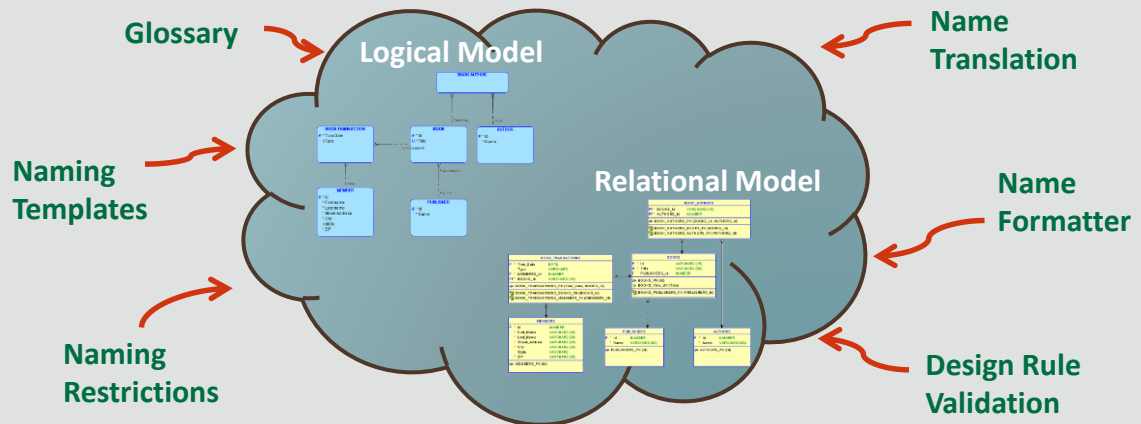
Naming Restrictions with Oracle Database

- Table and column names:
 - Must start with a letter
 - May contain up to 30 alphanumeric characters
 - Must not contain spaces or some special characters
 - Must avoid reserved words
- Table names must be unique within a schema
- Column names must be unique within a table

Each RDBMS can have its own naming restrictions. You must know whether the convention you decide to use is compatible with the RDBMS that you choose.

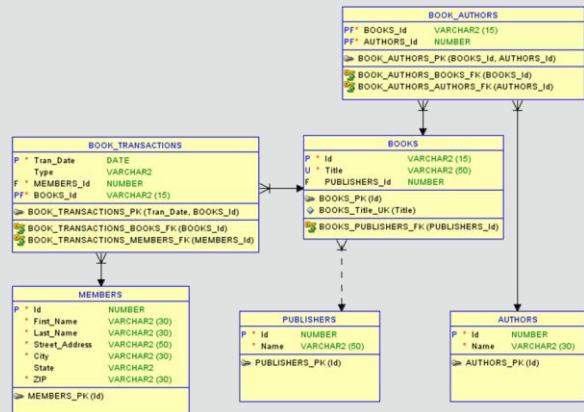
Applying Naming Standards Using Oracle SQL Developer Data Modeler

- There are many ways to define naming standards in Oracle SQL Developer Data Modeler. In this lesson we will create a Glossary to map entities to table names, and attributes to column names



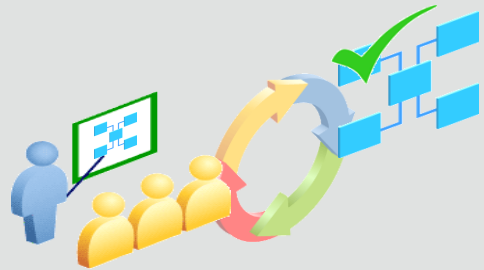
Applying Naming Standards Using Oracle SQL Developer Data Modeler (cont.)

- The previous lesson engineered from a Logical to a Relational Model. By default, table names are mapped directly from entity names, and are not plurals
- Column names are mapped directly from attribute names, with the exception that spaces are replaced by underscores
- To apply our naming conventions to Table and column names, we create a glossary



Glossary

- A glossary in Data Modeler is a set of accepted terms that can be used in the design
- It ensures that the model complies with your set naming standards or when engineering between the logical and relational models
- You can create a new glossary, use existing glossaries, or generate a glossary based on an existing logical model



ORACLE
Academy

DFo 5-1
Mapping Entities and Attributes

Copyright © 2020, Oracle and/or its affiliates. All rights reserved.

10

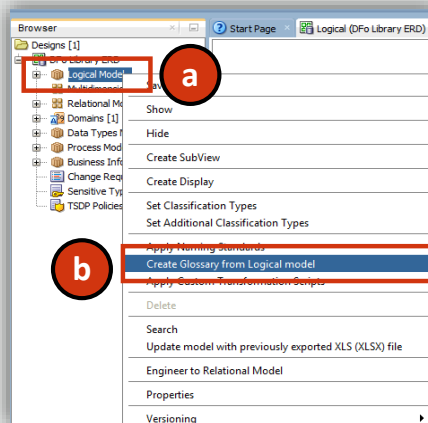
After you create a Glossary, an XML file is created with a .glossary extension. To modify the glossary, you must use the Glossary Editor and not an external text editor.

Create a Glossary from a Logical Model

- You can create a glossary from the existing logical model. This is quicker and easier than creating a new glossary, as entity and attribute names from the logical model are added automatically

Create a Glossary from a Logical Model

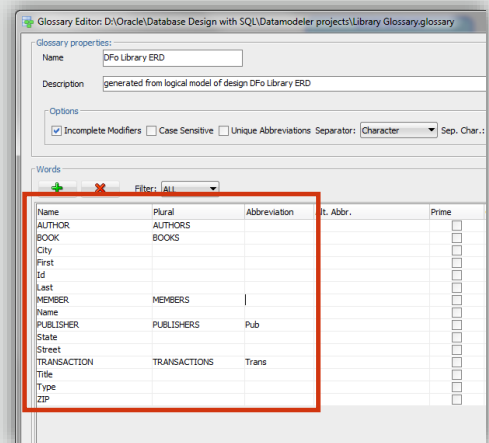
- To create a glossary from an existing logical model:
 - a. Right mouse click on the Logical model in the browser window
 - b. Select Create Glossary from Logical Model



Creating a
Glossary from
a Logical Model

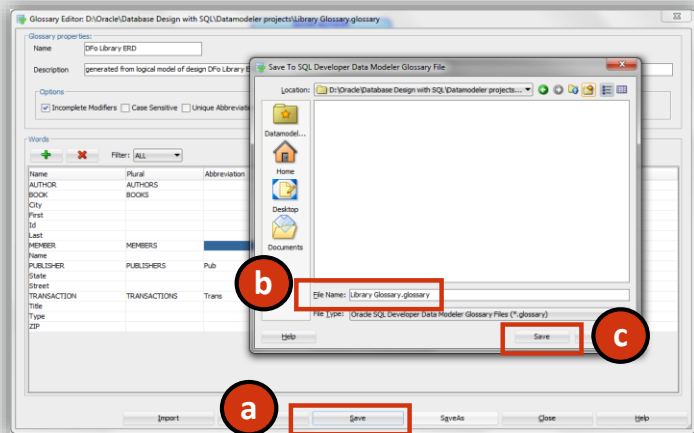
Create a Glossary from a Logical Model

- The Glossary editor includes suggested plurals and abbreviations. Edits can be made by clicking into the field and modifying or deleting the text
- Ensure Entity names have plurals, and any required abbreviations are included for attributes. Remove any suggestions you do not require



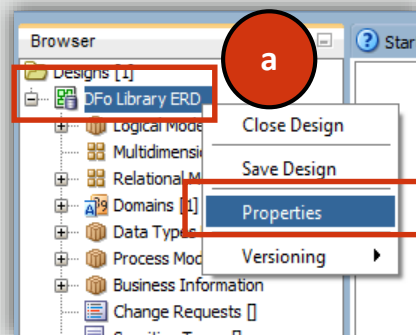
Create a Glossary from a Logical Model

- When you have made all the required changes
 - a. Click save
 - b. Enter a filename for the Glossary and browse to a location to save the file (Note the location for use in the next step)
 - c. Click save



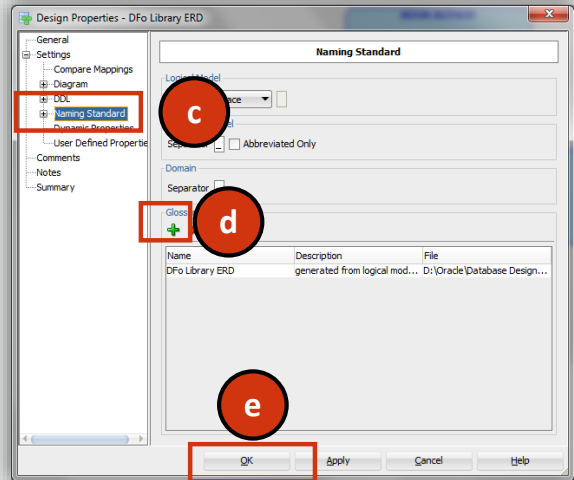
Adding the Glossary as the Naming Standard

- For the glossary to be applied during engineering, you must add it on the Naming Standard page in the Preferences dialog box.
- To add the glossary, perform the following steps:
 - a. Right-click the Design model in the Browser
 - b. Select Properties



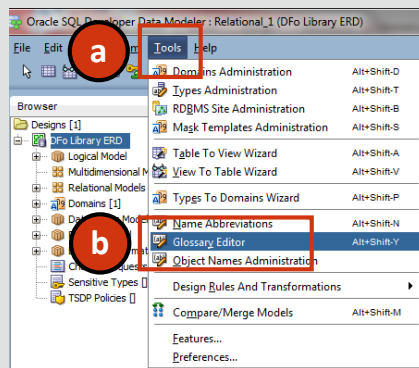
Adding the Glossary as the Naming Standard

- To add the glossary, perform the following steps:
 - c. Expand Settings and highlight the Naming Standard node
 - d. Click the green + icon, and navigate to the location of the glossary
 - e. Click OK



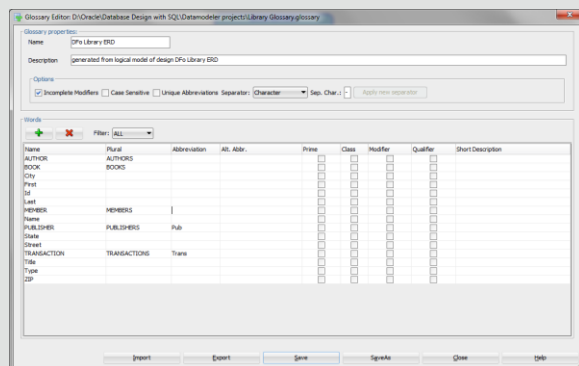
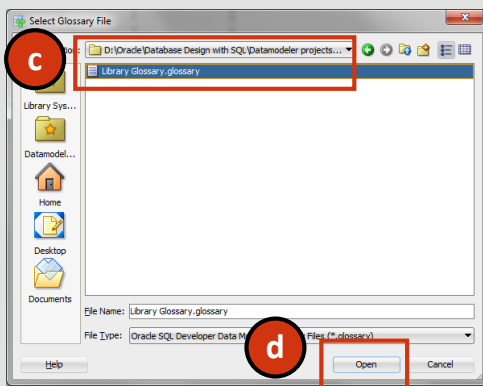
Glossary Editor

- To edit an existing glossary
 - a. Click Tools
 - b. Click Glossary Editor



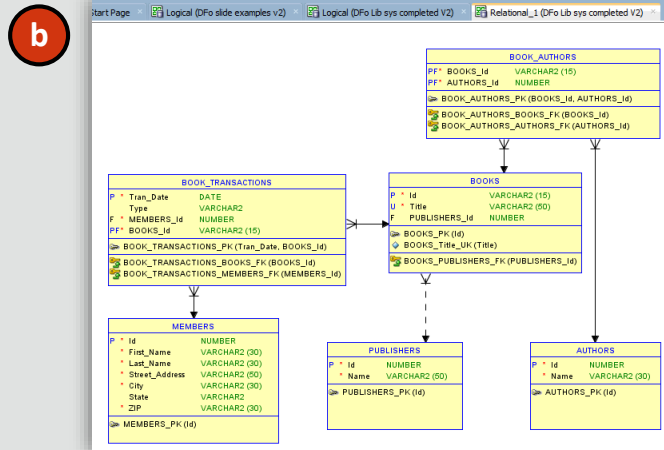
Glossary Editor

- To edit an existing glossary
 - c. Browse to the location of the required glossary file, and click
 - d. Select Open



Applying the Naming Standard

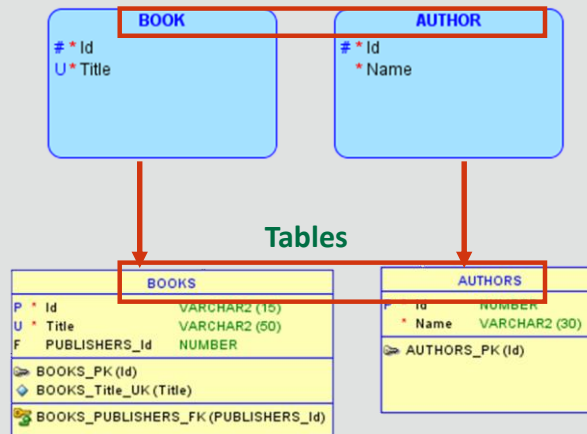
- To Apply the Glossary as the naming standard,
 - a. Engineer the Logical model once more
 - b. Click the relational tab to view the results



Mapping Entities to Table names

- Entities

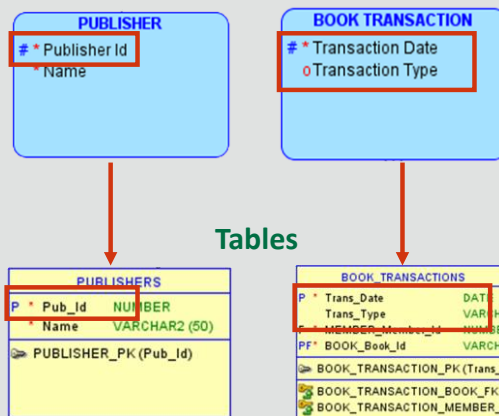
- By applying the naming standards contained in the glossary, singular entity names in the logical model are mapped to plural table names in the relational model



Mapping Attributes to Column names

- Entities

- If, for example, we had included the terms Publisher and Transaction in our Entity attribute names, by applying the naming standards contained in the glossary, the abbreviations specified for Publisher and Transaction are applied in the column names



Summary

- In this lesson, you should have learned how to:
 - Explain the naming conventions used in a relational database
 - Use Oracle SQL Developer Data Modeler to create a glossary and apply naming standards to
 - Map entities to table names
 - Map attributes to column names



The logo for Oracle Academy is centered on a light gray background. It features the word "ORACLE" in a bold, orange, sans-serif font. Below it, the word "Academy" is written in a smaller, dark gray, sans-serif font. The entire logo is framed by two horizontal dark gray bars, one at the top and one at the bottom.

ORACLE

Academy